ARUSHA REGION

PHYSICS MARKING SCHEME

Mock F4. 2026 March

1.

i	ii	iii	iv	v	vi	vii	viii	ix	x
B	C	E	D	D	B	D	D	E	B

2.

i	ii	iii	iv	v	vi
F	B	G	A	H	E

3. @ - Rules to be considered when constructing a ray diagram in a Concave mirror:

i/ A ray of light parallel to the principale axis, should be reflected through the principal focus (F.) (01)

ii/ A ray of light passing through the principale focus should be reflected parallel to the principal axis. (01)

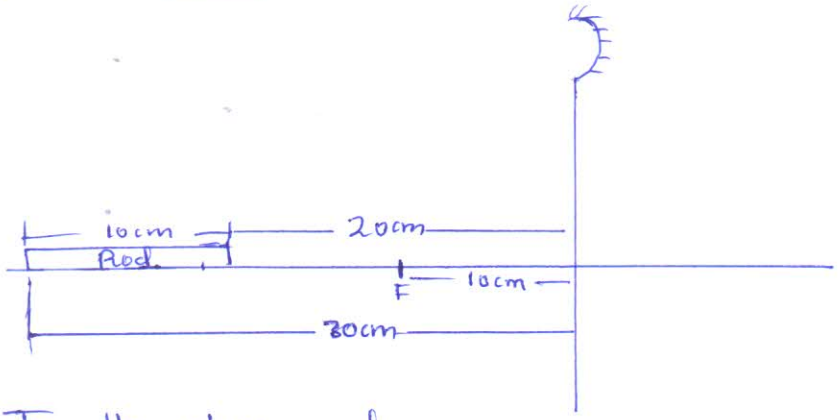
iii/ A ray of light passing through the centre of the curvature should be reflected through its own path or the same path. (01)

iv/ A ray of light passing through the pole of the mirror, should pass unreflected. (01)

01@ = 04 mark

01

3. (b) Soln:



For the closer end,

Focal length (f) = 10cm

object distance (u_c) = 20cm

image distance (v_c) = required

From;

$$\frac{1}{f} = \frac{1}{u_c} + \frac{1}{v} \quad (01)$$

$$\frac{1}{v_c} = \frac{1}{f} - \frac{1}{u_c}$$

$$v_c = \frac{uf}{u-f}$$

$$v_c = \frac{20\text{cm} \times 10\text{cm}}{(20\text{cm} - 10\text{cm})}$$

$$v_c = 20\text{cm} \quad (01)$$

For the for end

Focal length (f) = 10cm

object distance (u_f) = 30cm

Image distance (v_f) = required

From; $\frac{1}{f} = \frac{1}{u_f} + \frac{1}{v_f}$

3 (b) $\frac{1}{f} = \frac{1}{u_f} + \frac{1}{v_f}$ (01)

$\frac{1}{v_f} = \frac{1}{f} - \frac{1}{u}$

$v_f = \frac{uf}{u-f}$

$v_f = \frac{30\text{cm} \times 10\text{cm}}{(30\text{cm} - 10\text{cm})}$ (01)

$v_f = 15\text{ cm}$

∴ Length of the image $= v_c - v_f$

$= 20\text{cm} - 15\text{cm}$

$= 5\text{ cm}$ (01)

Image length (L) = 5 cm

4. (a) From the principle Conservation of linear momentum, total momentum before collision must be equal to total momentum after collision if there is no external force acting on a colliding system.

Momentum before collision = Momentum after collision

Momentum before collision = 0

Momentum after collision $= M_g V_g - M_b V_b$

$M_g V_g - M_b V_b = 0$

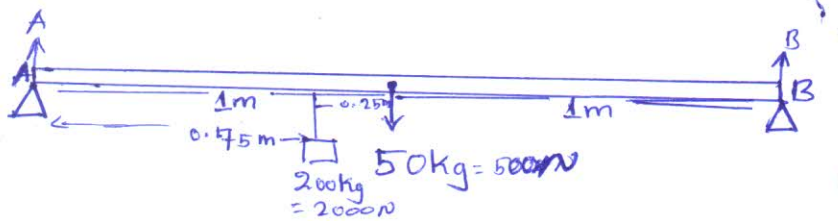
$M_g V_g = M_b V_b$

03

4. @ $\frac{V_g}{V_b} = \frac{M_b}{M_g}$

From the equation above, Since mass of the bullet is much less Compared to mass of the gun, hence the recoil velocity of the gun is also much less than the velocity of the bullet. (04)

(b) Consider the diagram below:



(01)

Take moment at A

From the Principle of moment;

Sum of C.M = Sum of A.M (01)

$(2000 \times 0.75) + (500 \times 1) = (B \times 2)$

$1500 + 500 = 2B$

$\frac{2000}{2} = \frac{2B}{2}$

$B = 1000N$

$B = 1000N$ (01)

4. (b) From:

Sum of upward forces = Sum of downward forces

$A + B = 2000N + 500N$

$A + 1000N = 2500N$

$A = 2500N - 1000N$

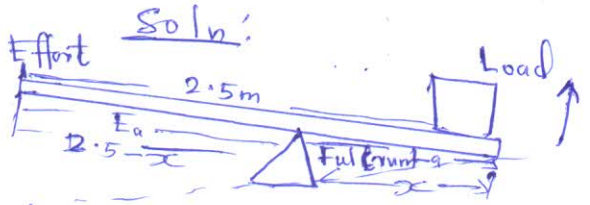
$A = 15000N$ (01)

Thrusts at A and B are 1500N and 1000N Respectively (01)

5. (a) i/ This is because of the friction force which acts between the machine parts. (02)

ii/ Because of the energy lost in term of sound and heat. (02)

(b) i/ Soln:



Data:

Velocity Ratio (V.R) = 12

Efficiency Eff = 80 %

Effort arm (Ea) = Required

Load (L) = 750

05

5. (b) From:

$$\text{Eff} = \frac{M.A}{V.R} \times 100\% \quad (01\frac{1}{2})$$

$$M.A = \frac{V.R \times \text{Eff}}{100\%}$$

$$M.A = \frac{12 \times 80\%}{100\%}$$

$$M.A = 9.6 \quad (01\frac{1}{2})$$

Again;

$$M.A = \frac{E_a}{L_a} \quad (01\frac{1}{2})$$

let $L_a = x$,

$E_a = (2.5 - x)$

$$M.A = \frac{(2.5 - x)}{x}$$

$$9.6 = \frac{(2.5 - x)}{x}$$

$$9.6x = 2.5 - x$$

$$\frac{10.6x}{10.6} = \frac{2.5}{10.6}$$

$$x = 0.24m \quad (01)$$

Now;

$$E_a = 2.5m - 0.24m$$

$$E_a = 2.26m \quad (01)$$

The effort is applied 2.26m from the fulcrum.

06

5 (b) ii/ From;

$$M.A = \frac{L}{E}$$

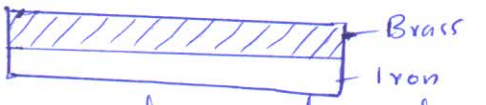
$$E = \frac{L}{M.A}$$ 01½

$$E = \frac{95\ N}{9.6}$$

$$E = 7.8\ N$$ 01

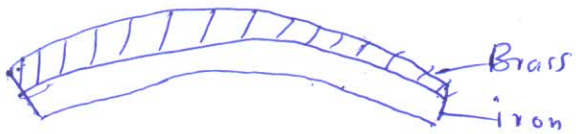
∴ The Effort required is 7.8 N

6. (a) A bimetalic strip made of iron and brass.



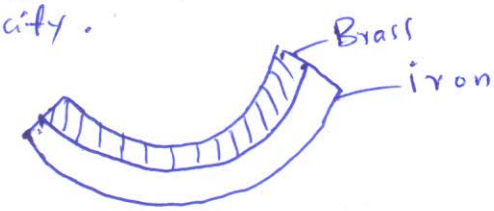
at room temperature

→ A bimetalic strip when heated it will bend toward iron because brass has high specific heat capacity. Compared to iron so brass expand fast or more than iron.



02

→ When cooled, it will bend toward brass because brass cool fast than iron because brass has high specific heat capacity.



02

07

6 (b) Data given :

Mass of cold water (m_c) = 215g = 0.215kg

Mass of hot water (m_h) = 105g = 0.105kg

Temperature of hot water (θ_h) = 75 °C

Temperature of cold water (θ_c) = 15 °C

Specific heat capacity of water = 4200 J/kg °C.

Final temperature of the mixture (θ_f) = required.

From :

Heat lost by hot water = Heat gained by cold water

$$m_h C_w \Delta\theta_h = m_c C \Delta\theta_c \quad (01)$$

$$m_h c_w (\theta_h - \theta_f) = m_c c_w (\theta_f - \theta_c)$$

$$m_h\theta_h - m_h\theta_f = m_c\theta_f - m_c\theta_c \quad (01)$$

$$m_h\theta_h + m_c\theta_c = m_c\theta_f + m_h\theta_f$$

$$\theta_f (m_c + m_h) = m_h\theta_h + m_c\theta_c$$

$$\frac{\theta_f (m_c + m_h)}{(m_c + m_h)} = \frac{m_h\theta_h + m_c\theta_c}{(m_c + m_h)}$$

$$\theta_f = \frac{m_h\theta_h + m_c\theta_c}{(m_c + m_h)} \quad (01)$$

$$\theta_f = \frac{(0.105 \times 75) + (0.215 \times 15)}{(0.215 + 0.105)}$$

$$\theta_f = 34.7\ °C \quad (02)$$

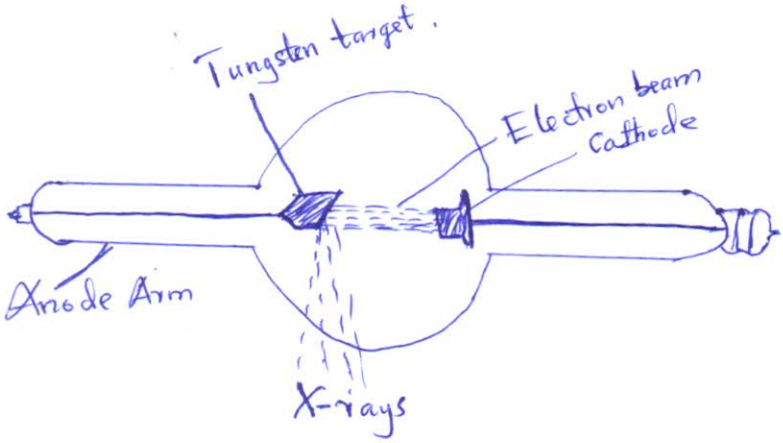
∴ The final temperature of the mixture is 34.7 °C

08

7. (a) i/ Hard X-rays have high penetrating power while soft X-rays have low penetrating power. (02)

ii/ Hard X-rays have very high energy while soft X-rays have low energy. (02)

(b) X-rays are produced when the fast moving electrons are stopped by a metal target as shown below;



(05)

X-Ray tube

8. (a) - Venus' size is almost as big as the Earth.
- Venus density is the same as that of the Earth
- Venus mass is the same as that of the Earth
- Venus internal structure resemble with that of the earth. Any two (02) @ = 04 marks

8. (b) i/ Two negative effects of volcanic activity on the Earth Surface.

- Loss of life and properties
- Environmental pollution (02½)

ii/ Two possitive Effects of volcanic activitie on the Earth Surface.

- Formation of landscapes
- Soil fertility. (02½)

9. (a) Factors which affect resistance of a Conductor

- Length of the Conductor (01)
- Temperature. (01)
- Area of cross-section of the Conductor (01)
- Nature of the material of the Conductor (01)

(b) Data given

Resistance (R) = 100 Ω

Potential difference (V) = 240V

i/ Current (I) = ?

11 (c) Data given

Primary voltage (V_P) = 240V

Secondary voltage (V_S) = 12V

Primary turns (N_P) = 600 turns

Secondary turns (N_S) = Required!

From;

$$\frac{V_P}{V_S} = \frac{N_P}{N_S} \quad (01)$$

$$N_S = \frac{V_S N_P}{V_P} \quad (01)$$

$$N_S = \frac{12V \times 600}{240V} \quad (01)$$

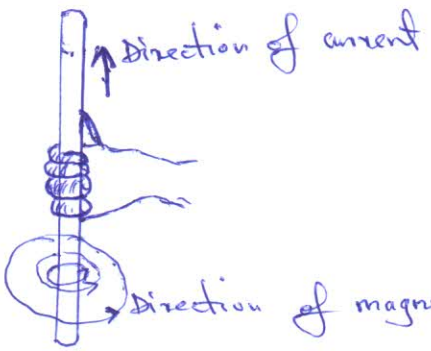
$$N_S = 30 \text{ turns}$$

(03)

Number of turns in the secondary coil is 30 turns

11. (a) From gripping right-hand rule.

"When wrap your right hand around a current carrying Conductor, the wrapping finger will point the direction of magnetic field and the thumb will point the direction of current." (02)



(02)

(b) i/ Evidence of existance of magnetic field around a current carrying Conductor.

→ A magnetic compass needle brought near to a current carrying Conductor will deflect. (02)

(Deflection of a magnetic Compass needle when brought near a current carrying Conductor)

ii/ - Increasing current in the wire (01)
- Increasing the number of turns of the coil (01)
- Using Soft iron core. (01)

10. (c) iii/ The perpendicular distance (h)

Distance (d) = 2d

Time (t) = 0.6$ + 0.25$

= 0.85 $

From;

$$v = \frac{2d}{t}$$ (01½)

$$d = \frac{vt}{2}$$

$$d = \frac{333.3 \times 0.85}{2}$$

$$d = 141.67\ m$$ (01½)

From;

$$a^2 + b^2 = c^2$$

Now; where, $a = 100\ m$ (01½)

$b = h$

$c = d = 141.67\ m$

$$b^2 = c^2 - a^2$$ (01½)

$$h^2 = (141.67)^2 - (100^2)$$

$$h = \sqrt{(141.67)^2 - (100)^2}$$

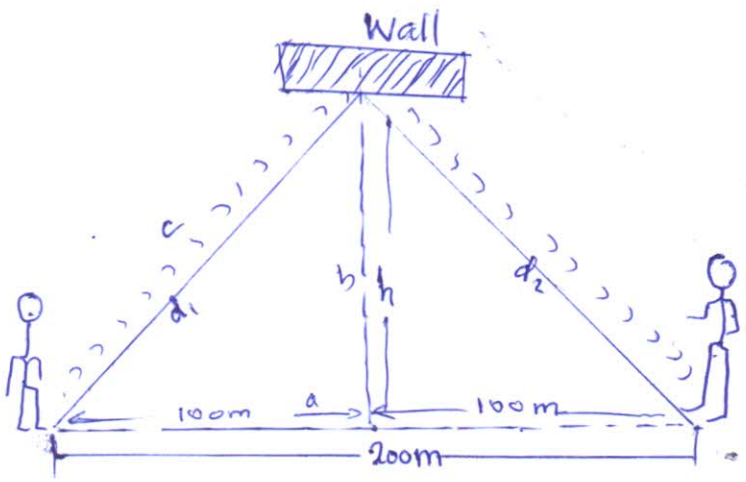
Perpendicular distance $(h) = 100.4\ m$ (01)

10. (b) iii/ Mass per unit length of the string.

- The thicker and heavier a string is, the lower its frequency for a given length and tension. (02)

$$f \propto \frac{1}{\sqrt{\mu}}$$

(c) i/



(01)

The diagram showing the position of the persons.

ii/ The velocity of Sound (V)

Data given:

Distance (d) = 200m

Time (t) = 0.6s

Velocity (V) = required.

From:

$$V = \frac{d}{t} \quad (01\frac{1}{2})$$

$$V = \frac{200m}{0.6s} \quad (00\frac{1}{2})$$

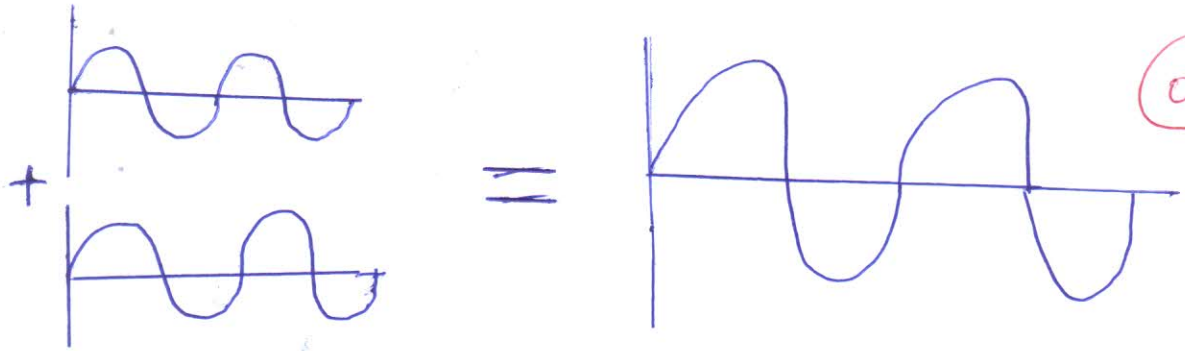
$$V = 333.3\ m/s$$

(01)

∴ The velocity of Sound (V) = 333.3m/s

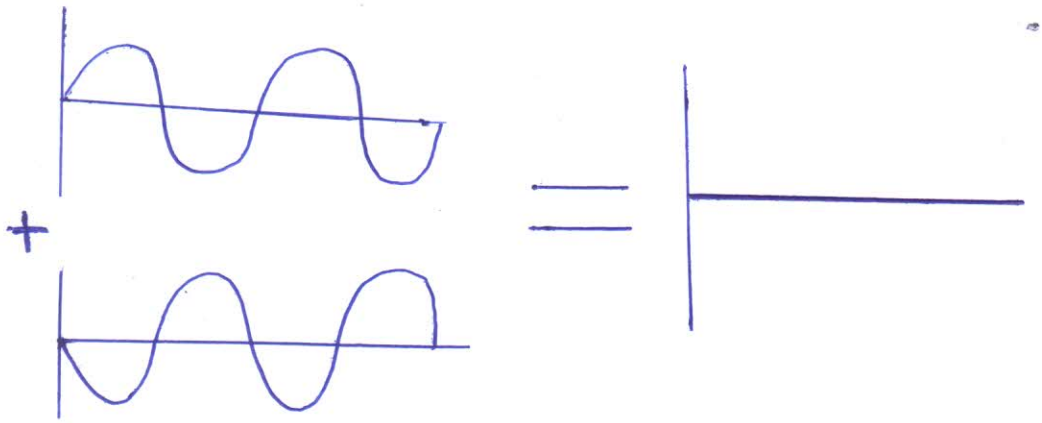
14

10. (a) i/ Constructive interference occurs when two equal waves travelling in the same direction meets or Supperpose.



02

ii/ Distructive interference occurs when two equal waves travelling in opposite direction meet or Supperpose.



02

(b) i/ By changing its length, when the length of the string is changed, it will vibrate with the different frequency. The shorter the string the higher the frequency and the longer the string the lower the frequency.

OR

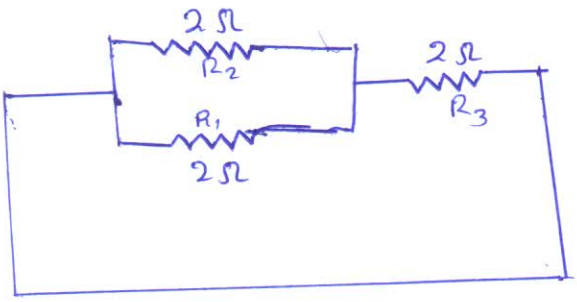
$$f \propto \frac{1}{L}$$

ii/ By changing its Tension, tightening a string gives it a higher frequency while loosening it lower the frequency. $f \propto \sqrt{T}$.

02

9. (c) The technician should connect two resistors in parallel and one resistor in series as shown.

02



01

Total resistance

$$R_T = \frac{R_1 R_2}{R_1 + R_2} + R_3$$ 01

$$R_T = \frac{(2 \times 2)\Omega}{(2+2)\Omega} + 2\Omega$$

$$R_T = \frac{4\Omega}{4\Omega} + 2\Omega$$

$$R_T = 1\Omega + 2\Omega$$ 01

$$R_T = 3\Omega$$

9 (b) i/ From;

$$V = IR$$ (01)

$$I = \frac{V}{R}$$

$$I = \frac{240V}{100\,\Omega}$$

$$I = 2.4\,A$$ (01)

$\therefore$ Current flowing (I) = 2.4 A

ii/ Power (P)

From: $P = IV$ (01)

$$P = 2.4 \times 240$$

$$P = 576 \text{ Watts}$$ (01)

$\therefore$ Power rating (P) = 576 Watts

iii/ Energy (E)

From: $E = P \times t$ (01)

$$E = 576 \times (8 \times 60 \times 60)$$

$$E = 16,588,800\,J$$ (01)

$\therefore$ Energy E = 16,588,800 J